Multi-Step Proportion Problems

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lessons 10–11 — Multi-Step Proportion Problems	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.AR.4.5

2. Standards

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

Must. Solve a two- or three-step proportional problem (a rate used twice; a recipe scaled and checked; a part-to-whole mixture), and compare two rates after putting them in one unit with a printed reference-sheet ratio. (MA.7.AR.4.5, with MA.7.AR.3.3 inside it.)

Must not. A graph, a table of a proportional relationship, or y = px named as such (MA.7.AR.4.1–4.4, Unit 7); a ratio not on the FAST sheet; a percent problem (Unit 8).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.1.1	Actively participate in effortful learning both individually and collectively.	Notes I and board 4: a problem whose first answer (miles per tank) is not the answer, and the room keeps going
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Notes II and board 6: 'which is faster?' is refused until the units match; the answer is checked against the estimate that a mile is longer
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	boards 8–9: a plan's area in square meters and a choice of car — the unit's whole toolkit on one real question

4. Learning Target and Essential Question

Learning target. I can solve a problem that takes two or more proportional steps, including a unit conversion, and compare two rates given in different units.

Essential question. A car gets 100 km/h and another does 65 mi/h. Why can't you say which is faster yet?

Building on. 5.06–09 — the sheet's ratios; 5.04 — a scale, then a speed: two rates in one problem; Grade 6 — a unit rate.

Working toward. Unit 7 — the constant of proportionality as a graph and an equation; the Unit 5 test's two transfer items, which are boards 8 and 9 in a third context.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A car goes 402.5 mi on a tank. How many tanks does a 1,207-mi trip take, and why is the answer a whole number?	DOK 2
Two rates are 12 km in 50 min and 7 mi in 45 min. Why does 12 > 7 tell you nothing yet?	DOK 3
Paint is mixed 2 : 4 : 5. If you know the white, how do you find the whole batch — and why is that a scale factor?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A car gets 28 miles per gallon. Its tank holds 15 gal. How far does it go on a full tank?	***420 mi***	1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]; 43 — added
2	A printer prints 35 pages per minute. A job is 1,400 pages. How long does the job take?	***40 min***	49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]; 400 — place value
3	Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days. How much does she earn?	***$435***	87 — one day's pay [Notes I: the question asks for five days]; 2,175 — 435 × 5, the days counted twice
4	A car gets 30 miles per gallon with a 12-gal tank. It makes a 900-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.)	***D — 3 tanks***	(A) 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]; (B) 900 ÷ 12: the gallons used as miles per tank [Notes I: the rate first]; (C) 900 ÷ 36: the two numbers added before dividing [7.AR.4.5: the steps not named — Notes I]
5	A cereal has 12 g of protein per 100 g. How many grams of protein in 1 kg?	***120 g***	12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]; 1,200 — × 100
6	$1\,\text{mi} = 1.61\,\text{km}$. Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min. Who is faster?	***C — cyclist B***	(A) 12 km and 12 km/h confused: a distance read as a speed [7.AR.4.5: a rate is per unit of time — Notes II]; (B) 12 > 7, the distances compared in two units [Notes II: same unit first]; (D) 11.27 is B's distance in km, not a speed — the time was never used [7.AR.4.5: a rate is per unit of time — Notes II]
7	Paint is mixed white : blue : grey = $2:4:5$. A batch uses 8 gal of white. How many gallons is the whole batch?	***44 gal***	20 — the grey (5 × 4) and nothing more [Notes III: the whole is all the parts]; 32 — 8 × k, the white scaled again
8	A plan is drawn at $1\,\text{in} : 4\,\text{ft}$; a room is 2 in by 2.5 in on it. $1\,\text{ft} = 0.305\,\text{m}$. Find the room's area in square meters.	***About 7.44 m²***	24.4 — 80 ft² × 0.305, the ratio once on an area [5.08–09 Notes II]; 20 — 2 × 2.5 × 4, the drawing's area scaled once [5.04 board 6]
9	$1\,\text{mi} = 1.61\,\text{km}$. A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h. Which car gets there first, and by how long? Explain.	***Car A: 80.5 km/h against 75, so about 3.73 h against 4 h — about 16 minutes sooner.***	'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]; 'Car A, 80.5 > 75' with no times — the decision without the 'by how long'

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Every board starts with the unit rate written as 'per 1' before anything is multiplied. For a comparison, a two-column table — one unit across the top — stays on the board, and nothing is compared until both columns are filled.

ELL. **Per** means for each one; **rate** is a per-sentence (miles per gallon, pages per minute). **Tank** on a board means a full tank of fuel.

Enrichment. Board 7 finishes early for some: ask for the batch if the grey were 25 gal instead ($k = 5$, 55 gal), and why the whole is always $11 \times k$.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the two speeds in one unit (50 mi/h is 80.5 km/h), the comparison, and the decision in a sentence with its numbers. A board that picked a car without converting has compared numbers, not speeds.

Vocabulary. **unit rate**; **proportional**; **per**

Materials. Whiteboards and markers. The FAST reference sheet's conversion lines on the board (printed on every item as well). The Unit Review key to post. Calculators are allowed on everything and printed on nothing. 43 minutes: the timings below are the short ones.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Compare rates: word problems — VVS; Multi-step problems with customary or metric unit conversions — UCT.